NON-EFFECTIVE FINITE EQUIVALENCE RELATIONS ON FINITE SCHEMES OVER A FIELD: VENKEN'S EXAMPLE CHECKED, AND A SECOND ONE

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Abstract. In 1961 Grothendieck asked whether a closed equivalence relation on a finite scheme over a field, with finite but not necessarily flat projections, is always effective. In a lecture at Buffalo in 1973 he still called the question open. It had been answered in the negative in 1971 by Venken, in a two-page note about the descent of flat modules whose Corollary 3.1 is a non-effective equivalence relation on the spectrum of a local Artin algebra of dimension 33 in six variables, over any field. We recheck Venken's example by linear algebra: the ideal of his relation exceeds, by one dimension in degree 4, the ideal of $X \times_Y X$, where Y is the quotient. We then give a second example in two variables, of length 15 (16 in characteristic 2), obtained by restricting Raicu's non-effective relation on the affine plane to a finite subscheme cut out by squares of invariant functions. For this one the comparison is carried out over $\mathbb{Q}$, the primes dividing a pivot are recorded, and those characteristics are checked separately, so that it holds over every field. We also prove it by hand, in every characteristic at once, with a linear form on the degree-5 part of $A \otimes_C A$, where C is the subalgebra generated by the three invariant functions.

1. The question

Let k be a field, $X = \operatorname{Spec} A$ a finite k-scheme and $R \subset X \times_k X$ a closed subscheme whose points with values in every k-algebra form an equivalence relation. Its projections $R \rightrightarrows X$ are finite, but they need not be flat. The ring of invariants

$$B = A^R = \{\, a \in A : a \otimes 1 - 1 \otimes a \in I_R \,\},$$

with $I_R \subset A \otimes_k A$ the ideal of R, is a finite k-algebra, and $Y = \operatorname{Spec} B$ is the quotient of X by R in the category of schemes. The relation is *effective* when the morphism $R \to X \times_Y X$ is an isomorphism.

In the third of his talks on the construction of schemes, after the flat case, Grothendieck wrote that without flatness he did not know whether a finite equivalence relation is effective. He reduced the question to the spectrum of a local Artin ring with algebraically closed residue field, and said that it was open even over a field [3, §2, p. 212-05]. The flat case is SGA 3, exp. V [8].

The answer is in print. Venken's note of 1971 [9], presented to the Académie on 5 April 1971, is titled after the descent of flat modules, and most of it is about modules: it constructs a finite injective morphism f of local Artin rings which is a descent morphism for flat modules, and a descent datum on a free module which is not effective. On the way, its Corollary 3.1 states that a certain ideal of $B \otimes_A B$ is an equivalence ideal and that the equivalence relation it defines is not effective (his B is our A, and his A a subring of it). The ring is a local Artin algebra over an arbitrary field K, with residue field K, so this is a negative answer to the question over every field. Venken takes his definitions from Grothendieck's first talk on descent [2].

Two years later, at the end of a lecture of his algebraic geometry course at the State University of New York at Buffalo, on 11 July 1973, a listener asked for open problems and

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he gave this one: he said it was not known, that he expected the answer to be negative, and that it was very elementary [1]. In the appendix to Kollár's paper on quotients by finite equivalence relations, Raicu gave finite equivalence relations on $\mathbb{A}^2$ which are not effective [4, §50], and affine toric equivalence relations are effective [6]. Philippe studied descent along unramified morphisms [5].

This note does two things. Section 3 checks Venken's Corollary 3.1 by machine, since his note gives it in one line after a cohomological computation. Sections 4 and 5 give a second example, smaller and in two variables, from Raicu's relation.

Theorem 1 (Venken, 1971)**.** *Let K be any field. There is a local Artin K-algebra of dimension 33, with residue field K, and a closed equivalence relation on its spectrum, with finite projections, which is not effective.*

Theorem 2. *Let k be any field. There is a graded local Artin k-algebra of dimension 15 (16 if k has characteristic 2) with two generators, with residue field k, and a closed equivalence relation on its spectrum, with finite projections, which is not effective.*

Both examples also answer the more general question, for X finite over a noetherian base.

2. Effectivity as linear algebra

With the notation of the question, $X \times_Y X = \operatorname{Spec}(A \otimes_B A)$ and $A \otimes_B A = (A \otimes_k A)/J_B$, where J_B is the ideal generated by the $b \otimes 1 - 1 \otimes b$, $b \in B$. By definition of B, $J_B \subseteq I_R$, and $R \to X \times_Y X$ is the closed immersion defined by I_R/J_B. So R is effective if and only if $I_R = J_B$, a comparison of two subspaces of the finite-dimensional space $A \otimes_k A$.

When A and I_R are graded, B is a graded subalgebra and both ideals are graded, so the comparison can be made degree by degree. A homogeneous invariant of degree e contributes to J_B only in degrees $\geq e$; to compare the two ideals in degree d it suffices to know the invariants of degree at most d.

3. Venken's example

Let K be a field. In Venken's notation [9], $\mathbf{B} = K[M, N, X, Y, Z, W]/\mathbf{I}$, where $\mathbf{I}$ is generated by

$$M^2,\ MN,\ N^2,\ X^3,\ Y^3,\ Z^3,\ W^3,\ YZ,\ NY,\ MZ,\ NXY,\ MYW$$

and all monomials of degree 3 in X, Y, Z, W; lower-case letters denote the classes. Let $\mathbf{A} \subset \mathbf{B}$ be the subalgebra generated by m, n, mx, nw and $my - nz$. Its maximal ideal has square zero, so $\dim_K \mathbf{A} = 6$. Venken proves that $\mathbf{A} \to \mathbf{B}$ is a descent morphism for flat modules, that $H^1(\mathbf{B}/\mathbf{A}, \mathbb{G}_a) \neq 0$, with $mxy \otimes w$ giving a nonzero class, and deduces:

> *Corollary 3.1.* The ideal $\mathbf{J}$ of $\mathbf{B} \otimes_{\mathbf{A}} \mathbf{B}$ generated by $mxy \otimes w$ is an equivalence ideal, and the equivalence relation it defines is not effective.

As a closed subscheme of $\operatorname{Spec}(\mathbf{B} \otimes_K \mathbf{B})$, this relation R_V has ideal

$$I_V = (a \otimes 1 - 1 \otimes a : a = m, n, mx, nw, my - nz) + (mxy \otimes w),$$

since the first generators cut $\mathbf{B} \otimes_{\mathbf{A}} \mathbf{B}$ out of $\mathbf{B} \otimes_K \mathbf{B}$. Its points are an equivalence relation on $\operatorname{Spec} \mathbf{B}$ over K, and its projections are finite because $\mathbf{B}$ is. All the generators are monomials or binomials of one degree, so the setting of Section 2 applies with every variable of degree 1.

Proposition 3. *Over every field K:*

1. $\dim \mathbf{B} = 33$*, and* $\mathbf{B}$ *vanishes in degrees above* 3*;*
2. I_V *satisfies the three axioms of an equivalence relation: it lies in the kernel of the multiplication, it is stable under the exchange of factors, and its image under p_{13} lies in the sum of its images under p_{12} and p_{23} in $\mathbf{B}^{\otimes 3}$;*
3. *the ring of invariants $\mathbf{B}^{R_V}$ has dimension* 10*, so it is strictly larger than* $\mathbf{A}$*;*
4. I_V *and* $J_{\mathbf{B}^{R_V}}$ *differ only in degree* 4*, where their dimensions are* 269 *and* 268*, and $mxy \otimes w$ does not lie in $J_{\mathbf{B}^{R_V}}$.*

So R_V is not effective in the sense of Section 1, as Venken states. Venken notes that the ideal generated by M^2, MN, N^2 and the monomials of degree 3 in X, Y, Z, W would suffice, the other generators only simplifying his computations. The same check on that ideal gives $\dim \mathbf{B} = 45$, invariants of dimension 8, and again a gap of one dimension in degree 4 ($410 > 409$).

The script `venken.py` of the directory `computations/effective/venken` of the repository [7] works in the monomial basis of $\mathbf{B}^{\otimes 2}$ and $\mathbf{B}^{\otimes 3}$ and performs the comparisons by Gaussian elimination over $\mathbb{Q}$. Every pivot it meets is ± 1, so the elimination runs over $\mathbb{Z}$ and gives the same ranks and kernels over every prime field, hence over every field; it was also run directly over $\mathbb{F}_p$ for $p = 2, 3, 5, 7, 11, 101$, with the same result.

4. Raicu's relation and its restriction

Write $x = (x_1, x_2)$ and $y = (y_1, y_2)$ for the coordinates of $\mathbb{A}^2 \times \mathbb{A}^2$. Raicu's relation [4, §50] is the closed subscheme $R \subset \mathbb{A}^2 \times \mathbb{A}^2$ with ideal

$$I = (x_1^2 - y_1^2,\ x_1x_2 - x_2^2 - y_1y_2 + y_2^2,\ x_2^3 - y_2^3,\ (x_1y_2 - x_2y_1)\, y_2^3) \subset k[x_1, x_2, y_1, y_2].$$

It is a finite equivalence relation on $\mathbb{A}^2$, defined over $\mathbb{Z}$, and it is not effective after base change to any field; the class of $(x_1y_2 - x_2y_1)\, y_2^3$ is the obstruction. The functions

$$f_1 = x_1^2, \qquad f_2 = x_1x_2 - x_2^2, \qquad f_3 = x_2^3$$

are invariant, since $f_i(x) - f_i(y)$ is one of the first three generators of I. Let

$$Z = V(f_1^2, f_2^2, f_3^2) \subset \mathbb{A}^2, \qquad A = k[x_1, x_2]/(f_1^2, f_2^2, f_3^2).$$

Since $f_1^2 = x_1^4$ and $f_3^2 = x_2^6$, Z is supported at the origin, and A is a local Artin algebra with residue field k; a count of monomials shows that the Hilbert function of A is $1, 2, 3, 4, 3, 2$, so that $\dim_k A = 15$. In characteristic 2, $f_2^2 = x_1^2x_2^2 + x_2^4$ loses its middle term, the Hilbert function is $1, 2, 3, 4, 3, 2, 1$ and $\dim_k A = 16$; the proof by hand in Section 5.1 uses only degrees at most 5. Let $R_Z = R \cap (Z \times Z)$, the closed subscheme of $Z \times Z$ with ideal I_{R_Z}, the image of I in $A \otimes_k A$.

Lemma 4. *R_Z is a closed equivalence relation on Z with finite projections.*

Proof. For every k-algebra T, $R_Z(T) = R(T) \cap (Z(T) \times Z(T))$ is the restriction of an equivalence relation on $\mathbb{A}^2(T)$ to the subset $Z(T)$, hence an equivalence relation. R_Z is closed in $Z \times Z$, which is finite over Z by either projection. □

The restriction to Z is what lets a question about $\mathbb{A}^2$ be asked about a finite scheme. It is not formal that non-effectivity survives it: the ring of invariants of R_Z is computed on A, not on $k[x_1, x_2]$, and it can be larger than the image of the invariants on $\mathbb{A}^2$, so that J_B can grow. The next section shows that it does not grow enough.

5. The computation

All the ideals above are homogeneous for the standard grading. In $k[x, y] = k[x_1, x_2, y_1, y_2]$ put $K = (f_1^2, f_2^2, f_3^2)(x) + (f_1^2, f_2^2, f_3^2)(y)$, so that $A \otimes_k A = k[x, y]/K$. The ideal of R_Z lifts to $I + K$, and J_B lifts to $J + K$, where J is generated by the $b(x) - b(y)$ for b running over homogeneous lifts of a basis of B.

Proposition 5. *Over $\mathbb{Q}$ and over every finite prime field, the invariants of R_Z of degree at most 7 give 27 independent differences $b(x) - b(y)$ modulo $I + K$, and in the degrees $d \leq 14$ the two ideals differ only in degree 5, where*

$$\dim_k (I + K)_5 = 52 > 51 = \dim_k (J + K)_5.$$

By the remark at the end of Section 2, invariants of degree at least 8 do not change $J + K$ in degree 5, so the strict inequality in degree 5 proves that R_Z is not effective. A witness is the generator $(x_1y_2 - x_2y_1)\, y_2^3$ of I, Raicu's obstruction: it lies in $(I + K)_5$ and not in $(J + K)_5$, as we checked over $\mathbb{Q}$, $\mathbb{F}_2$ and $\mathbb{F}_3$.

The proposition is a finite computation: the degree-d parts are spanned by monomial multiples of the generators, the invariants of degree d are the kernel of a linear map on the forms of degree d in x, and the dimensions are ranks. The script `scan.py` of the directory `computations/effective/raicu` of the repository [7], run with `EXP=2,2,2`, performs it by Gaussian elimination over $\mathbb{Q}$ or over $\mathbb{F}_p$.

Proof that it holds over every field. A field k contains $\mathbb{Q}$ or some $\mathbb{F}_p$, and all the data are defined over the prime field: ranks and kernels of matrices with entries in the prime field do not change under extension of scalars. It remains to cover every prime field.

Over $\mathbb{Q}$ the elimination divides only by pivots. Each coefficient it produces is an integer combination of the input coefficients, all integers, divided by a product of pivots. If a prime p divides neither the numerator nor the denominator of any pivot, the same sequence of operations is defined over $\mathbb{F}_p$, with the same pivot positions, and gives the same ranks and the same kernels. The script `certify_scan.py` reruns the computation over $\mathbb{Q}$ and records the prime factors of all pivots; they are $2, 3, 5, 7, 13, 41$ and 61. For every other prime the proposition over $\mathbb{F}_p$ follows from the one over $\mathbb{Q}$. For these seven primes `scan.py` was run directly over $\mathbb{F}_p$, with the same result. □

Remark 6. Why this Z. Among the subschemes $V(f_1^a, f_2^b, f_3^c)$ with $a, b, c \leq 3$, sixteen were tried over $\mathbb{F}_3$, and the smallest on which the obstruction survives is $a = b = c = 2$; with exponents $(3, 3, 2)$ one gets length 27, with gaps in degrees 5 and 6. Whether a shorter example exists, from other ideals, is not known. A random search over 48 small monomial algebras over $\mathbb{F}_2$ and $\mathbb{F}_3$ found no non-effective relation; Raicu's theorem that affine toric equivalence relations are effective [6] suggests why. Compared with Venken's example, this one has length 15 (or 16) against 33 and two generators against six.

Remark 7. Grothendieck reduced the question to an algebraically closed residue field; both examples have residue field k, which may be chosen algebraically closed. Both are also graded, which the question does not require.

5.1. **A proof by hand.** We now prove by hand that $w = (x_1y_2 - x_2y_1)\, y_2^3$ lies in $(I + K)_5$ and not in $(J+K)_5$, over every field and without computing B. Let $C \subset A$ be the subalgebra generated by the classes of f_1, f_2, f_3; it lies in B. Put $g_i = f_i(x) - f_i(y)$, the first three generators of I, and

$$L = (g_1, g_2, g_3) + K \subset k[x, y],$$

so that $k[x, y]/L = A \otimes_C A$ and $L \subseteq J + K \subseteq I + K$.

Some degrees of A. In degrees at most 3, A coincides with $k[x_1, x_2]$. In degree 4 the relations are $x_1^4 = 0$ and $f_2^2 = x_1^2x_2^2 - 2x_1x_2^3 + x_2^4 = 0$, so A_4 has dimension 3, with basis $x_1^3x_2$, $x_1^2x_2^2$, $x_1x_2^3$, and

$$x_2^4 = 2x_1x_2^3 - x_1^2x_2^2. \tag{1}$$

In degree 5, $x_1^5 = x_1^4x_2 = 0$, and $x_1f_2^2 = x_2f_2^2 = 0$ give $x_1x_2^4 = 2x_1^2x_2^3 - x_1^3x_2^2$ and $x_2^5 = 3x_1^2x_2^3 - 2x_1^3x_2^2$. So A_5 is spanned by $x_1^3x_2^2$ and $x_1^2x_2^3$, and since $f_1f_3 = x_1^2x_2^3$ and $f_2f_3 = x_1x_2^4 - x_2^5 = x_1^3x_2^2 - x_1^2x_2^3$,

$$C_5 = A_5. \tag{2}$$

These formulas have integer coefficients and hold in every characteristic. The graded pieces of C in low degree are $C_1 = 0$, $C_2 = kf_1 + kf_2$, $C_3 = kf_3$, $C_4 = kf_1f_2$ (because $f_1^2 = f_2^2 = 0$ in A) and $C_5 = A_5$.

Lemma 8. *$(J + K)_5 = L_5$, and $(I + K)_5 = L_5 + kw$.*

Proof. The second equality holds because w is the only generator of I of degree 5 and the others are g_1, g_2, g_3. For the first, $L \subseteq J + K$ since f_1, f_2, f_3 are invariant. Conversely, J_5 is spanned by the products $(b(x)-b(y))\, m$ with b a homogeneous invariant of degree e, $1 \leq e \leq 5$, and m a monomial of degree $5-e$. If $e \leq 4$, then $b(x) - b(y)$ lies in $(I+K)_e$, which equals L_e because w has degree 5; so the product lies in L. If $e = 5$, (2) gives a polynomial c in f_1, f_2, f_3

with $b - c \in (f_1^2, f_2^2, f_3^2)$; then $b(x) - b(y)$ is congruent modulo K to $c(x) - c(y)$, which lies in (g_1, g_2, g_3). □

So R_Z is not effective as soon as $w \notin L_5$, that is, as soon as the class of w in $(A \otimes_C A)_5$ is not zero. The invariants of R_Z play no further role. The lemma also accounts for the dimensions of Proposition 5: $52 = 51 + 1$.

A presentation of $A \otimes_C A$. In $A/C^+A = k[x_1, x_2]/(f_1, f_2, f_3)$ one has $x_1^2 = 0$ and $x_1x_2 = x_2^2$, hence $x_1^2x_2 = 0$ and $x_1x_2^2 = x_2^3 = 0$; this ring has basis $1, x_1, x_2, x_2^2$. By the graded Nakayama lemma,

$$e_0 = 1, \quad e_1 = x_1, \quad e_2 = x_2, \quad e_3 = x_2^2$$

generate A as a C-module. Let $E \subset C^4$ be the module of relations, the (c_0, c_1, c_2, c_3) with $\sum c_j e_j = 0$ in A, graded by $\deg c_j + \deg e_j$. Tensoring $0 \to E \to C^4 \to A \to 0$ with A over C gives

$$A \otimes_C A = A^4/AE,$$

where (a_0, a_1, a_2, a_3) stands for $\sum a_j \otimes e_j$. In degree 5, $(A^4)_5 = A_5 \oplus A_4 \oplus A_4 \oplus A_3$, and $(AE)_5$ is spanned by the products ar with $r \in E_d$, $d \le 5$, and $a \in A_{5-d}$. The class of $w = x_1 \otimes x_2^4 - x_2 \otimes x_1x_2^3 = x_1 \otimes f_3e_2 - x_2 \otimes f_3e_1$ is

$$W = (0,\ -x_2^4,\ x_1x_2^3,\ 0).$$

The relations up to degree 5. A relation of degree d has $c_0 \in C_d$, $c_1, c_2 \in C_{d-1}$ and $c_3 \in C_{d-2}$.

- $E_d = 0$ for $d \le 2$: in degree 2 a relation reads $c_0 + \gamma x_2^2 = 0$ with $c_0 \in kf_1 + kf_2$ and $\gamma \in k$, and x_1^2, $x_1x_2 - x_2^2$, x_2^2 are independent; degrees 0 and 1 are clear.
- $E_3 = k\rho$ with $\rho = (f_3,\ f_2,\ f_2 - f_1,\ 0)$. The identity $f_3 + x_1f_2 + x_2(f_2 - f_1) = 0$ holds in $k[x_1, x_2]$. The five products f_3, x_1f_1, x_2f_1, x_1f_2, x_2f_2 span A_3 (they give x_2^3, x_1^3, $x_1^2x_2$, then $x_1x_2^2$), which has dimension 4, so the relations of degree 3 form a line.
- $E_4 = k\sigma_1 + k\sigma_2$ with

  $$\sigma_1 = (0,\ -f_3,\ f_3,\ f_2), \qquad \sigma_2 = (0,\ -2f_3,\ f_3,\ f_1).$$

  Indeed $-x_1f_3 + x_2f_3 + x_2^2f_2 = 0$ in $k[x_1, x_2]$, and $-2x_1f_3 + x_2f_3 + x_2^2f_1 = f_2^2$, which is 0 in A. The space $C_4 \oplus C_3 \oplus C_3 \oplus C_2$ has dimension at most 5, and its image contains $x_1f_3 = x_1x_2^3$, $x_2^2f_1 = x_1^2x_2^2$ and $f_1f_2 + x_2^2f_1 = x_1^3x_2$, which span A_4. So E_4 has dimension at most 2, and σ_1, σ_2 are independent (look at the last entries).
- An element of E_5 has $c_1 \in C_4 = kf_1f_2$ and $c_3 \in C_3 = kf_3$. Nothing more is needed.

The linear form. Let λ be the linear form on the forms of degree 4 with $\lambda(x_1^i x_2^{4-i}) = 1$ for $0 \le i \le 3$ and $\lambda(x_1^4) = 0$. Since $\lambda(f_2^2) = 1 - 2 + 1 = 0$, it is defined on A_4. Let ν be the linear form on A_3 with

$$\nu(x_1^3) = \nu(x_1^2x_2) = 2, \qquad \nu(x_1x_2^2) = 1, \qquad \nu(x_2^3) = 0,$$

and put $\varphi(a_0, a_1, a_2, a_3) = \lambda(a_1) + \nu(a_3)$ on $(A^4)_5$.

Lemma 9. *φ vanishes on $(AE)_5$, and $\varphi(W) = -1$.*

Proof. By the list above it suffices to evaluate φ on $a\rho$ for $a \in A_2$, on $b\sigma_1$ and $b\sigma_2$ for $b \in A_1$, and on E_5.

- $\varphi(a\rho) = \lambda(af_2)$, and for $a = x_1^2, x_1x_2, x_2^2$, af_2 is $x_1^3x_2 - x_1^2x_2^2$, $x_1^2x_2^2 - x_1x_2^3$, $x_1x_2^3 - x_2^4$: each has $\lambda = 1 - 1 = 0$.
- $\varphi(b\sigma_1) = -\lambda(bf_3) + \nu(bf_2)$. For $b = x_1$: $-\lambda(x_1x_2^3) + \nu(x_1^2x_2) - \nu(x_1x_2^2) = -1 + 2 - 1 = 0$. For $b = x_2$: $-\lambda(x_2^4) + \nu(x_1x_2^2) - \nu(x_2^3) = -1 + 1 - 0 = 0$.
- $\varphi(b\sigma_2) = -2\lambda(bf_3) + \nu(bf_1)$. For $b = x_1$: $-2 + \nu(x_1^3) = 0$. For $b = x_2$: $-2 + \nu(x_1^2x_2) = 0$.
- For $r \in E_5$, $\varphi(r) = \lambda(c_1) + \nu(c_3)$ is a combination of $\lambda(f_1f_2) = \lambda(x_1^3x_2) - \lambda(x_1^2x_2^2) = 0$ and $\nu(f_3) = \nu(x_2^3) = 0$.

Finally $\varphi(W) = -\lambda(x_2^4) = -1$. □

Proposition 10. *Over every field, $w \in (I+K)_5$ and $w \notin (J+K)_5$. Hence R_Z is not effective.*

Proof. By Lemma 9, φ induces a linear form on $(A \otimes_C A)_5 = (A^4)_5/(AE)_5$ that does not vanish on the class of w, so $w \notin L_5$. By Lemma 8, $(J + K)_5 = L_5$. □

All the numbers in the proof are integers, and the only one that has to be nonzero is $\varphi(W) = -1$; so the argument is the same in every characteristic, 2 and 3 included, although A has length 16 in characteristic 2. The proof does not determine B: it only uses that invariants of degree at most 4 give differences in L, and that $A_5 = C_5$. It does not transport Raicu's argument on $\mathbb{A}^2$ either; it compares R_Z with $Z \times_{\operatorname{Spec} C} Z$ directly. The script `hand.py` of the same directory checks each identity used above over $\mathbb{Z}$.

What is not done. Both examples are certified over every field by computer, and both have a proof by hand: Venken's in his note, the second in Section 5.1. Neither is formalised in a proof assistant, and the second has not been checked by a specialist of descent. The ring of invariants B of the second example is not determined by hand; the proof does not need it.

Use of language models. The question was located in the recording of the Buffalo lecture and in [3] with language models (Claude Opus 5.5, Anthropic), which also found Raicu's example, proposed the restriction to Z, wrote the scripts and a first draft of this text, under the author's direction. Venken's note was read on Gallica, and its Corollary 3.1 found to answer the question, after the first version of this text had claimed the case of a finite scheme over a field to be new. The proof by hand of Section 5.1 was found the same way, the scripts serving to search for the linear form φ; each of its steps is a short computation that a reader can redo with pencil and paper. Propositions 3 and 5 rest on the scripts, which anyone can rerun.

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